

Vaghesan Sundaram

 [github](#) |  [portfolio](#) |  [linkedin](#) |  vaghesan2@gmail.com

ABOUT ME

Computer Science (Cybersecurity track) and Mathematics undergraduate with hands-on experience in offensive security problem-solving, including CTFs and NSA Codebreaker Challenges, and production IT experience in a regulated environment. Strong foundation in Python scripting, network analysis, access control, and system security, with interest in penetration testing and enterprise risk reduction. Expected graduation: May 2027.

WORK EXPERIENCE

Community Assistant, University of Maryland

August 2025 – Present

- Provide service desk support, administrative assistance, and assist residents with diverse needs in a customer service environment.

New Jersey Courts – Ocean Vicinage — Summer Intern

June – August 2025

- Supported production IT systems in a regulated, high-availability environment, assisting staff across multiple departments.
- Managed user access, account provisioning, and workstation security, reinforcing least-privilege and secure configuration practices.
- Assisted with system updates, security patching, and configuration changes tied to compliance requirements.
- Collaborated on backup and recovery workflows, supporting data integrity and operational continuity.

Mentee, ZEB (AI Digital Transformation)

Dec 2023 – Mar 2024

- Gained exposure to AI-driven development workflows and intersections of AI and cybersecurity through structured mentorship covering model evaluation, experimentation, prompt engineering, secure software development, and data privacy.

CERTIFICATIONS

CompTIA Security+ (SY0-701)

March 2026 (Expires March 2029)

Covers network security, threat modeling, access control, incident response, risk management, and security operations.

PROJECTS & COMPETITIONS

NSA Codebreaker Challenge 2024 & 2025

Completed CTF-style challenges focused on incident response, reverse engineering, cryptography, and network analysis. Used Linux, scripting, debugging, and secure coding practices to analyze unfamiliar systems and behaviors.

Capture the Flag (CTF) Competitions - UMD Team

Participated in vulnerability analysis and exploitation challenges, including CSAW CTF Qualifications. Performed web, binary, and cryptography challenges, using Python scripts, reverse engineering tools, and network forensics techniques to identify and exploit weaknesses.

Keystroke Dynamics Authentication System

[GitHub](#)

Built a Python-based authentication prototype using behavioral biometrics. Designed a data pipeline for feature extraction, anomaly detection, and classification, exploring continuous authentication concepts.

Prophet Prediction Market - HackPrinceton 2025

[GitHub](#)

Developed backend components for an AI-driven prediction market on Solana. Integrated external AI agents with Python services and blockchain transactions, focusing on system design and data integrity.

TECHNICAL SKILLS

Cybersecurity & Networking Tools	Nmap, Wireshark, Burp Suite, OpenSSL, Linux security utilities, packet analysis, vulnerability assessment
Programming & Scripting	Python, Java, C++, x86-64 Assembly, Rust, Bash, OCaml
Systems & Software	Linux, Windows, Git/GitHub, debugging, unit testing, documentation
ML / Data	PyTorch, NumPy, scikit-learn, Hugging Face, Matplotlib

EDUCATION

University of Maryland, College Park	GPA: 3.700
B.S. Computer Science (Cybersecurity Focus)	Expected May 2027
B.S. Mathematics	Expected May 2027
Advanced Cybersecurity Experience for Students (ACES) – Cybersecurity Minor	
Middlesex College (MCC)	June 2024
Associate Degree in Mathematics	
Edison Academy Magnet School	June 2024
High School Diploma	

SELECTED COURSES

Discrete Mathematics • Linear Algebra • Algorithms • Computer Systems • Programming Languages • Differential Equations • Advanced Calculus I • Penetration Testing • Introduction to Abstract Algebra • Design and Analysis of Computer Algorithms • Computer and Network Security